Large Language Model-based agents for reconciliation of terms

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Abstract
The integration of diverse data across life sciences, healthcare, and related fields depends on the alignment of common terminology, which can be achieved by using ontologies and thesauri hosted in open repositories. However, mapping user-defined terms to labels from such repositories (e.g. Wikidata or BioPortal) remains a challenge due to the semantic ambiguity of concepts and constantly updating knowledge bases. The presented software is an open-source, Large Language Model (LLM)-assisted application designed to facilitate term reconciliation and mapping verification. The mapping process in our software relies on a semantic comparison between context-based definitions of input terms and endpoint-derived definitions of candidate labels. The software demonstrates how recent advances in LLM-driven agentic systems can be effectively leveraged to support efficient, standards-aligned terminology reconciliation, which remains a major bottleneck hindering the broad adoption of knowledge graphs in life sciences.

Keywords: Large Language Models, LLM-based agents, Agentic Systems, Semantic Web, Term reconciliation

Introduction

To avoid data silos and to increase the interpretability of data generated in life sciences, it is critical to make data truly interoperable. This can be achieved by linking data concepts and terms to cross-domain ontologies or thesauri hosted in publicly available knowledge graphs or repositories such as Wikidata[1] or BioPortal [2,3]. At the same time, finding the best-fitting concepts in these domains for a given term is a challenging process that usually requires tedious comparison of the term's semantic meaning and the candidate concepts. To facilitate this process, we have developed a multi-agent software system [4] that uses Large Language Models (LLM) and is specifically designed to reconcile terms against concepts from a predefined set of endpoints. The purpose of the system is to enrich available research data with ontological concepts semi-automatically, thereby supporting the development of a cross-domain knowledge graphs based on the Simple Knowledge Organisation System (SKOS)[5]. Furthermore, the system is proof of concept for the idea that Linked Data can be generated semi-automatically from any tabular research data. Below, we summarize the software design, while additional technical details and a ready-to-use application are presented in our open-source GitHub repository [4].

Technical description of the software

A multi-agent workflow is designed based on "Deep Agent" library from Langchain [6] and orchestrates the work of several LLM-based autonomous agents to reconcile terms used in datasets to entities in knowledge graphs and ontologies (Fig 1, left). Firstly, the dataset terms are enriched with definitions generated using a custom approach based on a Retrieval Augmented Generation (RAG) procedure[7]. The RAG database is built and queried with LightRAG[8] using the dataset description provided in an associated publication (e.g., a pdf). If a specific term from the dataset is not explicitly defined in the description, we run a few-shot LLM call that uses the description as context and RAG-derived term–definition pairs for dataset terms found in the description as examples. Secondly, specialized subagents use tools (e.g. specific functions) to communicate with APIs of the desired endpoints to (i) identify candidate labels through string and lexical matching and (ii) retrieve the corresponding definition for each candidate label. If the label is not directly linked to the definition, the

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definition is retrieved from a related label stored in the cross-ontology linkages. Finally, these specialized subagents perform a semantic comparison between the definition generated for the term and the definitions extracted from the candidate labels to decide which label has the "closest" match and then send the best candidate to the supervisor agent. To quantify the "closeness" between the definition of the term and the retrieved label the supervisor agent selects a matching class using SKOS-matching nomenclature (e.g skos:exactMatch, skos:closeMatch, skos:relatedMatch) [5]. The selection process is implemented with a custom-designed tool relying on a few-shot LLM call and structured output formatting [9]

Additionally, to allow the merging of our pipeline with the existing automatic services we show how the agents and tools can be transformed into individual Model Context Protocol (MCP) servers [10]. MCP servers enable independent execution of modules, thereby allowing users to construct custom workflows by composing agents, systems, and tools as reusable building blocks. The open-source nature of our design enables the end user to change the LLM version used, as well as to adapt the system prompts, the tool designs etc., thereby keeping the whole system up to date.

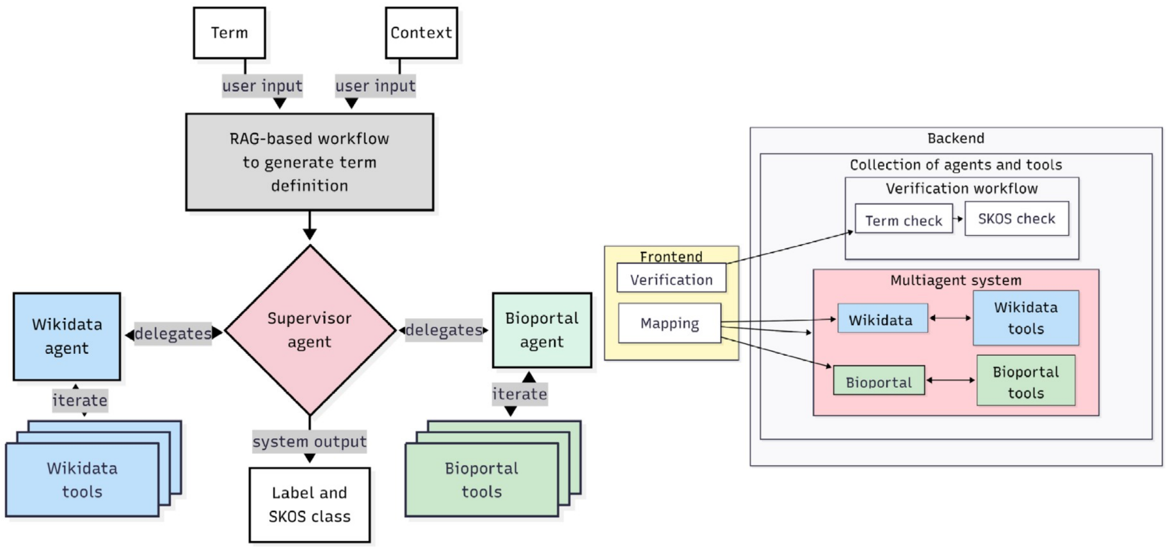

Figure 1: Left. Outline of the multi-agent system designed to map terms with labels in Wikidata and/or BioPortal endpoints based on the context-based definition of the term and endpoint-based definitions of the labels. More technical details, use cases and cost examples are in [4]. **Right.** Application design outlining endpoint mappings and mapping verification services.

The final application[4] consists of two separate services: mapping and verification (Fig 1, right). Within the mapping service the user provides a list of terms and chooses predefined endpoints of web-based resources with ontologies or knowledge graphs. Depending on whether one or several endpoints were chosen by the user, the system either redirects the mapping task to a single specialized deep agent or to a multi-agent system tailored to interact with the corresponding endpoints. For the verification service, the user provides a term along with the proposed label and the SKOS class. The system verifies the mapping and the SKOS classification by sequentially calling corresponding tools. The verification service design illustrates the use case in which an agentic approach is not required, but tools previously used by the agents are reused through a predefined workflow.

Conclusion and future directions

The proposed solution combines LLM-based autonomous agents with the Semantic Web to alleviate the bottleneck of reconciling thesauri and ontologies. Future work will add endpoints such as specialized NCBI resources [11] and integrate established mapping strategies as agent skills [12].

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Disclosure of Interests

The authors declare no competing interests relevant to the content of this article

Declaration on Generative AI

During the preparation of this submission, the authors used a Llama-3.3-70B Model provided within the KIDA project to check grammar, improve fluency and spelling of the text. The authors reviewed and edited the content as needed and take full responsibility for the publication's content.

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